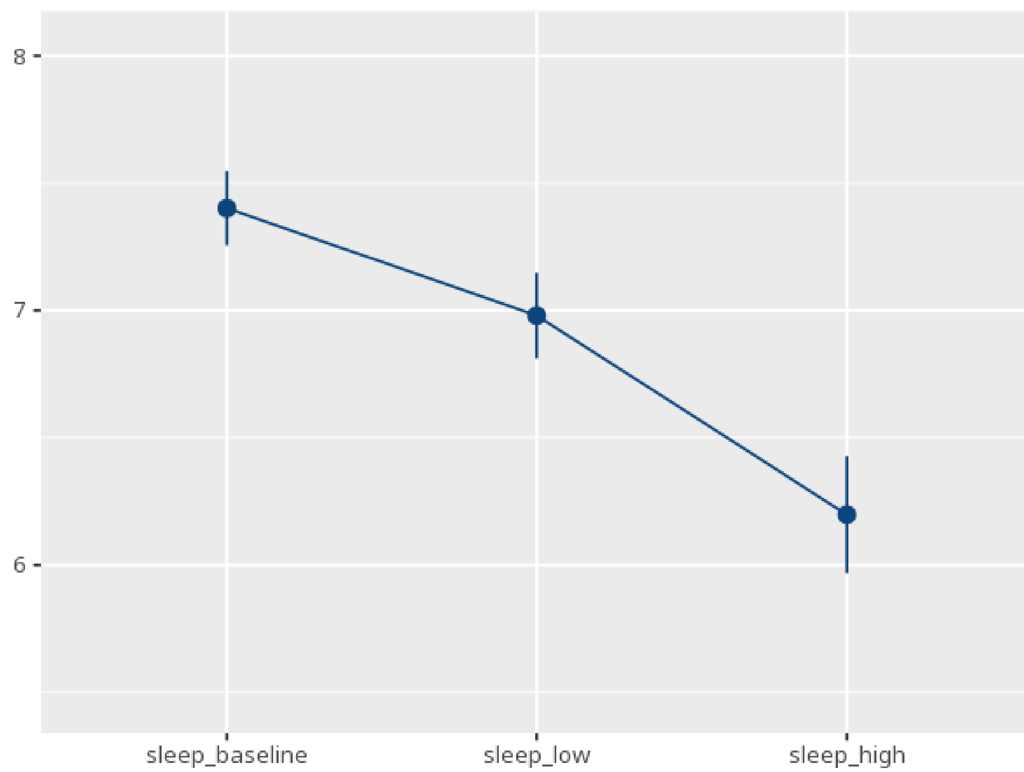
1 Repeated-measures ANOVA

Condition	N	M	SD
sleep_baseline	40	7.40	0.45
sleep_low	40	6.98	0.53
sleep_high	40	6.20	0.72

Source	SS	df	MS	F	p	partial η^2 [95% CI]
Condition	29.83	1.25	23.77	174.34	<.001	0.82 [0.76, 1.00]

Effect	W	p	GG ε	HF ε
condition	0.41	<.001	0.63	0.64

Pair	Mdiff	SE	padj	95% CI
sleep_baseline - sleep_low	0.42	0.04	<.001	[0.32, 0.52]
sleep_baseline - sleep_high	1.20	0.09	<.001	[0.99, 1.42]
sleep_low - sleep_high	0.78	0.06	<.001	[0.62, 0.94]



Tests whether the average differs across conditions measured on the same people. Check sphericity first - if violated, read the corrected F/p. A significant result means conditions differ; post-hoc tests show which. Your significance threshold (α) is 0.05.

A repeated-measures ANOVA (GG-corrected) gave $F(1.25,48.94)=174.34$, $p < .001$, partial $\eta^2=.82$ [.76, 1.00].

Statistical basis: Repeated-measures ANOVA (F-test for within-subjects conditions) (Fisher, R. A. (1925). Statistical Methods for Research Workers. Oliver & Boyd.); Mauchly's test of sphericity (rendered above the correction) (Mauchly, J. W. (1940). "Significance test for sphericity of a normal n-variate distribution." The Annals of Mathematical Statistics, 11(2), 204-209.); Greenhouse-Geisser correction (sphericity violated) (Greenhouse, S. W., Geisser, S. (1959). "On methods in the analysis of profile data." Psychometrika, 24(2), 95-112.); Huynh-Feldt correction (sphericity violated) (Huynh, H., Feldt, L. S. (1976). "Estimation of the Box correction for degrees of freedom from sample data in randomized block and split-plot designs." Journal of Educational Statistics, 1(1), 69-82.); Estimated marginal means for post-hoc comparisons (Lenth RV (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package.).